ENVS-193DS_homework-03

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Set up

```
library(tidyverse)
library(here)
library(janitor)
library(readxl)

kelp <- read_csv(here("data", "temp-kelp.csv"))
my_data <- read_xlsx(here("data", "Project 193DS.xlsx"))
```

Problems

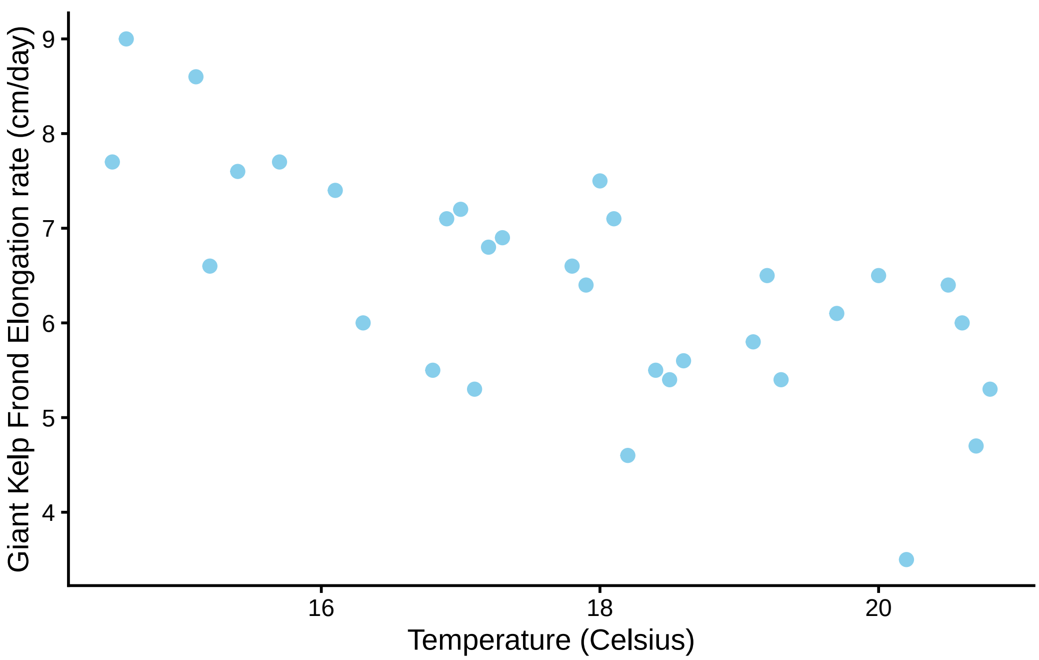
Problem 1. Giant kelp fronds

a. An appropriate test

An appropriate test to determine the strength of the relationship between temperature and giant kelp frond elongation rate would be to measure the correlation. You can measure stregth of correlation through the following tests: Pearson's product-moment correlation, Spearman's rank correlation.

b. Create a visualization

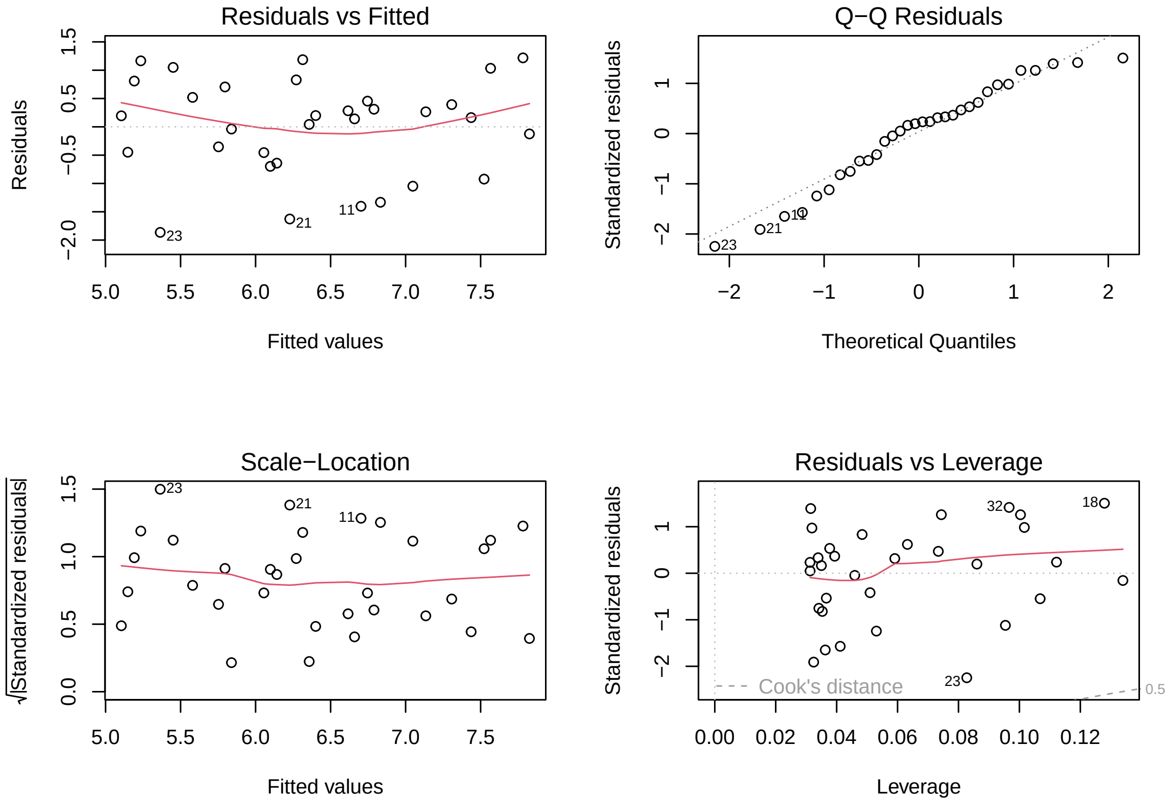
```r
# base layer ggplot
ggplot(data = kelp,
       # x axis = temperature
       # y axis = kelp_elong
       mapping = aes( x = temp_c,
                      y = kelp_elong)
       ) +
  # adding geometry - points, scatterplot
  geom_point(size = 2,
             # changing color from default
             color = "skyblue") +
  # changing theme from default
  theme_classic() +
  # adding axes labels
  labs( x = "Temperature (Celsius)",
        y = "Giant Kelp Frond Elongation rate (cm/day)")
```



c. Check your assumptions and run your test

Checking Assumptions

```r
# creating new object
# linear regression model of kelp
# (response ~ predictor)
kelp_lm <- lm( kelp_elong ~ temp_c,
    data = kelp
    )
#displaying linear model
# putting side by side
par(mfrow = c(2,2))
# plotting linear model
plot(kelp_lm)
```



For assumptions I wanted to check the homoscedasticity, normal distribution, and outliers of the model using a linear regression model. The outputs demonstrate random distribuions for Residuals vs Fitted and Scale-Location indicating homoscedasticity; points that, in general, match the qq line in the QQ plot, indicating normal distribution; and not really any outliers in the Residuals vs Leverage plot as the points lay inside of the dotted lines.

Running Correlation test

```r
# running pearsons coefficient
# data$predictor, data$response
cor.test(kelp$temp_c, kelp$kelp_elong,
         method = "pearson")
```

```

	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor 
-0.6877489 
```

d. Results communication

I used a Pearson's Correlation test in order to determine the strength of the relationship between temperature and giant kelp frond elongation rate. I found a moderate negative relationship between temperature and giant kelp frond elongation rate (Pearson's r = -0.69, t(30) = -5.2, p < 0.001, $\alpha = 0.05$).

e. Test implications

We found a moderate negative (Pearson's r = -0.69) relationship between temperature and giant kelp frond elongation rate. Here, as temperature increases, giant kelp frond elongation rate decreases.

f. Double check your own work

```
# double checking using Spearmans
cor.test(kelp$temp_c, kelp$kelp_elong,
         method = "spearman")
```

```
Warning in cor.test.default(kelp$temp_c, kelp$kelp_elong, method = "spearman"):
Cannot compute exact p-value with ties
```

```

    Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

After running the Spearman's rank test as well, I have concluded that the results lead me to the same conclusion of rejecting the null hypothesis for a moderate negative relationship between temperature and kelp frond elongation rate. Both tests have similar correlation values -0.68, along with p-values under 0.001.

Problem 2. Personal data

a. Updating your visualizations

Figure 1: Categorical Variable

```
# cleaning my data
# creating new object my_data_clean
my_data_clean <- clean_names(my_data) |>
  #making readable
  mutate(work_out = case_when(
    work_out == "N" ~ "No",
    work_out == "Y" ~ "Yes"
  ))
```

```r
#base layer: ggplot
ggplot(data = my_data_clean,
       #x- axis: did I work out the day before?
       #y- axis: ounces of coffee consumed per day
       mapping = aes(x = work_out,
                     y = coffee_oz,
                     #grouping colors by work out
                     color = work_out)) +
  #adding points
  geom_point(size = 2) +
  #labelling axes
  labs(x = "Work Out the Day Before?",
       y = "Coffee Per Day (oz)",
       title = "Influence of working out the day before on coffee consumed") +
  #choosing different colors, yes = green, no = red
  scale_color_manual(values = c("Yes" = "forestgreen",
                                "No" = "tomato")) +
  #changing theme from default and to make colors more visible
  theme_classic() +
  #getting rid of legend
  theme(legend.position = "none")
```

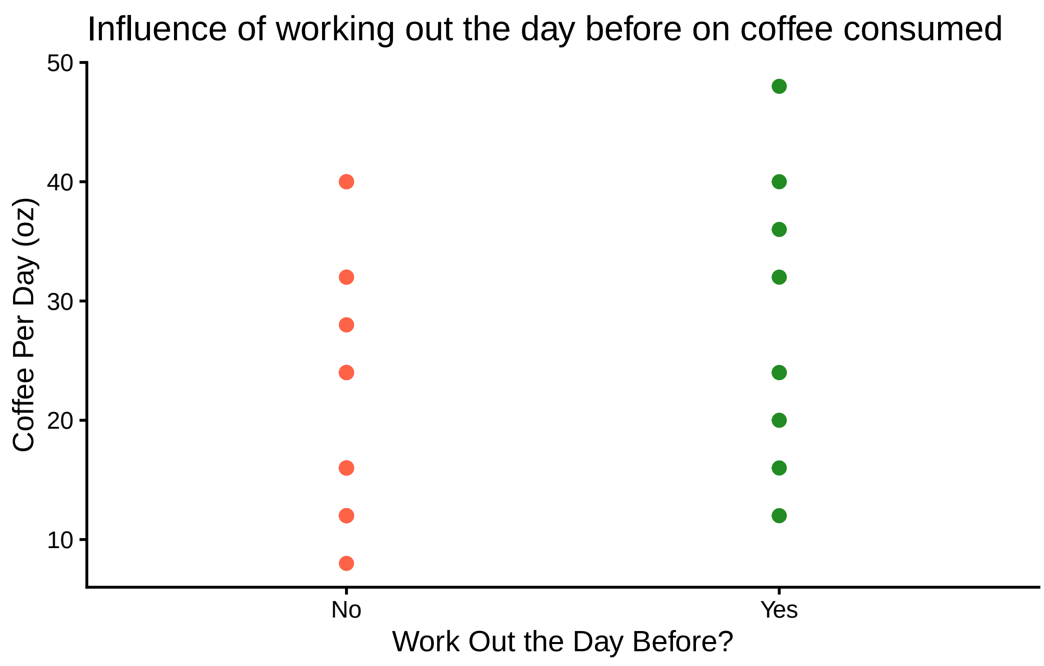
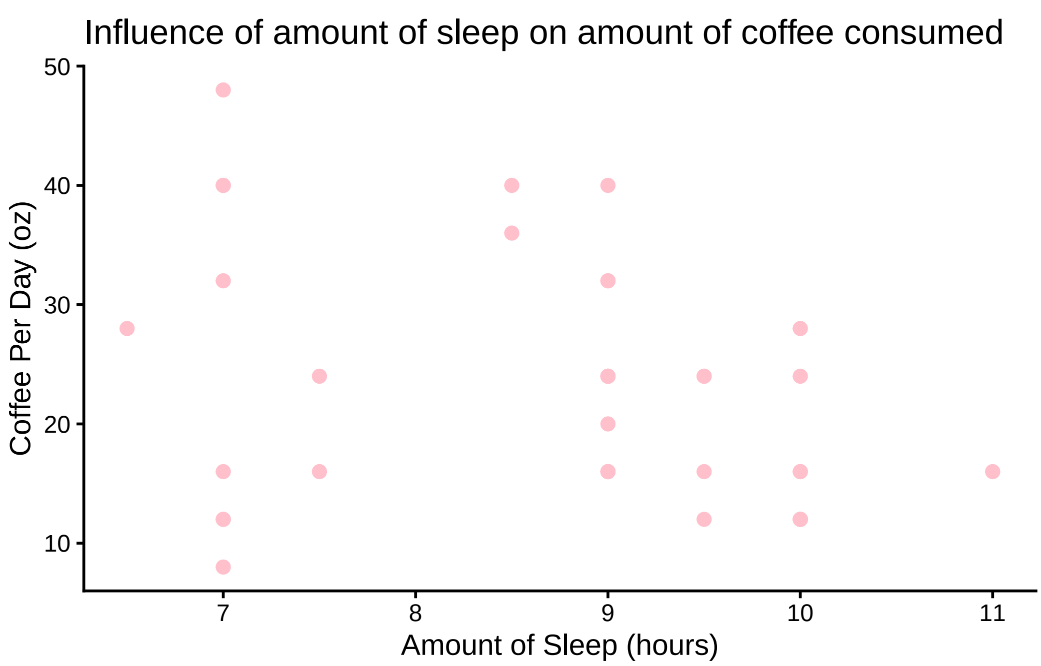


Figure 2: Continuous variable

```
#base layer: ggplot
ggplot(data = my_data_clean,
       #x- axis: Amount of sleep (hours)
       #y- axis: ounces of coffee consumed per day
       mapping = aes(x = amount_of_sleep,
                     y = coffee_oz
                     )) +
  #adding points
  geom_point(size = 2,
             color = "pink") +
  #labelling axes
  labs(x = "Amount of Sleep (hours)",
       y = "Coffee Per Day (oz)",
       title = "Influence of amount of sleep on amount of coffee consumed") +
  #changing theme from default
  theme_classic() +
  #getting rid of legend
  theme(legend.position = "none")
```



b. Captions

Figure 1: Working out the day before has little impact on coffee consumed per day (oz). Points represent observations of how much coffee, in ounces, was consumed depending on if I worked out the day before or not (total n = 37). Colors represent if I worked out the day before or not (green: Yes, red: No)

Figure 2: Coffee per day (oz) decreases as sleep (hrs) increases. Points represent observations of how much coffee, in ounces, was consumed compared to how many hours I had slept (n = 37).

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data

Because I like drinking coffee, I thought it would be nice to recreate my favorite way to drink coffee. Which is at a cafe or at home, cozy inside, with a great view of nature, mountains or otherwise, while I do my work. I was inspired from the watercolor mountain data, and I think it would be visually appealing to show my data as a mountain slope in the background of this coffee scene.

b. Create a sketch (on paper) of your idea

c. Make a draft of your visualization

d. Write an artist statement

e. Prep your materials to share in class

Problem 4. Statistical critique

a. Revisit and summarize

b. Visual clarity

c. Aesthetic clarity

d. Recommendations